Lab 2 – Product Specification

Team Bronze

Old Dominion University

CS411W

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Table of Contents

3. Product Requirements

3.1. Functional Requirements

3.1.1. User Interface (O: Jackie S.)

The system shall provide a clear and user-friendly interface for all members.

- The system will maintain a consistent layout and design across all pages on the website.
- The system shall allow users to easily navigate between all main pages
- The system will display relevant user and recall information in an organized manner.
- The system shall prioritize important recall information for quick user awareness based on their personalized account data.

3.1.1.1. Navigation Bar (O: Parker D.)

The system shall provide the user with a navigation bar at the top of the screen. This section of the website shall link to all accessible pages of the website. This section shall be shown on all pages of the website. The links shall be listed left to right in this approximate order:

- Home
- Recalls
- Pantry
- Settings
- Log In

3.1.1.2. Landing Page (O: Jackie)

- The system shall have a home page that welcomes the user, by name, to the website
- The system shall allow an easily usable navigation system
- The system shall give a dropdown menu that allows them to navigate to their personally added inventory, the recalls that apply to them based on said inventory, their settings, and a login menu
- The system shall provide the most important recall information first, immediately after login
- The system shall display the user's name to ensure they are logged into the correct account

3.1.1.3. Pantry Interface (O: Parker M.)

- The system shall provide an interface that enables users to view and manage the products added to their inventory.
- The system shall display a list of products submitted by the user.
- The system shall display the following information for each product:
  - Product name
  - Brand name
  - Universal Product Code (UPC)

3.1.1.3.1. Access Control (O: Parker M.)

- Upon accessing the pantry page, the system shall restrict access only to authenticated users.

- The system shall detect when an unauthenticated user attempts to access the pantry interface.
- The system shall display a prompt informing the user that login is required to access the pantry and provide the user with the option to navigate to either the home page or the login page.

3.1.1.3.2. Delete Items (O: Parker M.)

- The system shall provide the capability for the user to remove a purchased item from the user's inventory.
- The system shall permanently remove the deleted item from the set of products monitored for recall notifications.

3.1.1.3.3. Recall Information (O: Parker M.)

- The system shall provide a clear indication when a related recall exists for any product in the user's pantry.
- When a recall exists for a product in the user's pantry, the system shall provide all relevant recall information for that product.

3.1.1.3.4. Search and Filtering (O: Parker M.)

- The system shall provide a search capability that enables users to locate products within their pantry.
- The system shall allow users to search for products using the following attributes:
    - Product name

  - Brand name
  - UPC
- The system shall display a list of pantry items matching the user's input.
- The system shall allow users to clear an applied search and restore the full list of pantry items.

3.1.1.3.5. UPC Adding Input (O: Parker M.)

- The system shall provide an input field that enables users to add a product to their pantry by entering its UPC.
- The system shall validate the UPC entered by the user and display the corresponding product information in the user's pantry when a valid UPC is entered.
- The system shall notify the user if the entered UPC does not correspond to a recognized product.

3.1.1.4. Recalls Interface

3.1.1.4.1. Recall Feed (O: Mark S.)

- The system shall provide user specific recent recalls at the top of the page based on their set filters.
- The system shall provide the most recent FDA recalls below the personalized recalls.
- The system will allow users to select a recall to see more recall information in a drop-down format.

- The system shall provide a Class to each recall reflecting the severity of danger associated with the product in descending order, a lower class number being more dangerous.

3.1.1.4.2. Recall more info (O: Jackie S.)

- The system shall provide the user with specific information about recalls on products they have previously logged.
- The system will provide the name of the company as well as the specific product that was recalled, including the size of the container specifically contaminated.
- The system shall provide information on the reason for the recall (i.e., ingredients, allergens, etc.).
- The system will provide the user with the severity of the recall through its class, while also providing explanations of what each class means.
- The system shall give the user the region of the U.S. that is affected by this recall, by state.
- The system will report to the user the number of people affected by this recall, based on when the user sees it, as well as the number of deaths the recall has caused.

3.1.1.4.3. Search Recalls (O: Mark S.)

The system shall provide the user an interface to query the database for products by searching the product names and recalls reasons with the provided keywords.

- The system shall provide an input field that allows users to enter keywords on the Recalls page.
- The system will search product names based on users entered key words and display those products.
- The system will search product's recall reasons based on users entered key words and display those products.

3.1.1.4.4. Filter Recalls (O: Mark S.)

- The system shall provide an interface for selecting recall categories based on tags.
- The system shall toggle email based notifications based on selected tags.
- The system shall

3.1.1.5. Login Page (Parker M.)

The system shall provide a login page that enables registered users to access their accounts. The following functional requirements must be met:

- The system shall provide input fields for the user's email address and password.

- The system shall validate the entered credentials against stored account information and authenticate the user when valid credentials are provided.
- When authentication fails, the system shall display an error message to the user.
- The system shall provide a link to the registration page to allow new users to create an account.

3.1.1.6. Register Page (Parker M.)

- The system shall provide mandatory input fields for the user to enter the following information for account creation:
  - First name
  - Last name
  - Email address
  - Password
- The system shall prevent registration using an email address that is already associated with an existing account.
- The system shall create a new user account when valid registration information is submitted.

3.1.1.7. Settings Page (Shan I)

- The system shall allow the user to navigate to the Settings page from any other pages.

- The system shall display the user’s current profile information, including their First Name, Last Name, and registered Email address.
- The system shall allow the users to opt out of receiving information regarding recalled food.

3.1.1.8. Notifications (O: Shan I.)

- The system shall periodically monitor for active food safety recalls by querying a reliable external data source or internal database.
- The system shall automatically cross-reference the UPC/barcode and product details of newly issued recalls against the existing items currently stored in the user's virtual pantry.
- The system shall generate and dispatch a notification to the user only in the event of a direct match between a recalled product and an item actively logged in their pantry.
- The system shall utilize the Web Notification API to push real-time browser alerts to active users immediately when a recall match is detected in their inventory.
- The system shall simultaneously dispatch an automated email alert outlining the critical recall details (hazard type, product name, and recommended safety action) to ensure the user is informed even if they are not actively browsing the application.

- The system shall provide a clear link within the notification directing the user to the specific recalled item in their pantry dashboard so they can promptly remove or manage it.

3.1.2. Algorithms

3.1.2.1. Account Creation (O: Shan I., M: Joshua A.)

- The system shall display a registration form prompting the user for their First Name, Last Name, Email, and Password.
- The system shall validate that the email provided is not already associated with an existing user in the database.
- The system shall automatically generate a random 32 byte verification token and set the user's initial 'is_verified' status to 'false' upon form submission.
- The system shall automatically dispatch a verification email containing a unique activation link to the user's provided email address using the email delivery service specified in section 3.5.1..
- The system shall redirect the user to the Login page with a success message prompting them to check their email for the verification link.
- The system shall update the user's 'is_verified' status to 'true' and nullify the verification token only when the user clicks the valid activation link.

3.1.2.2. FDA Recall Grabbing (O: Parker D, M: Joshua A.)

The system shall read and access the external government recall data source specified in section 3.5.1 to routinely gather new official recall information. It shall request new recalls and parse all the needed information to store the recall in the database.

- The system shall retrieve updated recall information weekly from the external recall data source specified in section 3.5.1.
- The system shall take every individual recall and match it as closely as possible to an existing product in the database.
- For every new recall gathered during this process, an email will be sent out to relevant users as described in section **3.1.2.3**.
- Relevant information to be gathered:
    - Recall classification
    - Reason for recall
    - Number of people sick and injured
    - Product affected
    - UPC when available

3.1.2.3. Email Notification Sending (O: Parker D)

When a new recall is found according to the algorithm discussed in section **3.1.2.2**, an email is sent to all relevant users. Emails are sent to users whose notification filter settings allow them to receive this email.

- When a new recall is received, the system shall format an email to be sent to users. This email should include the same information as described in section **3.1.1.4.2**.
- The system shall check each individual user's notification filter settings before sending out emails. These settings can either allow all emails to be sent or only those that affect items in the user's pantry.

3.1.2.4. OpenFoodFacts Product Grabbing (O: Shan I., M: Joshua A.)

- The system shall provide an input field where users can enter a product's barcode (UPC) to add an item to their pantry.
- The system shall dispatch an HTTP request to the external product information data source specified in section 3.5.1 using the provided barcode as the query parameter.
- The system shall parse the JSON response from the external product information data source specified in section 3.5.1 to extract relevant product metadata, specifically the product name, brand, and image URL.
- The system shall automatically populate the "Add Item" form fields with the extracted metadata, allowing the user to verify the information before saving it to their database.
- The system shall display a clear error message if the external product information data source specified in section 3.5.1 does return a match for the entered barcode.

3.2. Non-Functional Requirements

3.2.1. Data Management and Security (O: Josh)

- The system shall require a unique email address for each registered user account and reject registration attempts using an email already associated with an existing account.
- The system shall hash all user passwords using bcrypt before storing them in the database. Plaintext passwords shall never be stored or logged.
- The system shall sanitize all user input prior to processing to prevent SQL injection and cross-site scripting (XSS) attacks.
- The system shall restrict each authenticated user's database access to records that match their own user ID, preventing users from accessing or modifying another user's data.
- The system shall remove all database entries associated with a user's account upon that user's request for account deletion.
- The system shall transmit all data between the client and server over HTTPS to protect data in transit.
- The system shall invalidate a user's session token upon logout, preventing reuse of expired session credentials.

3.3. Performance Requirements

3.3.1. Application Performance (O: Josh)

- The system shall load any page within three seconds under normal network conditions.
- The system shall respond to interactive user interface actions within two seconds.

- The prototype system shall support a minimum of 20 concurrent users without degradation in response time or functionality.
- The system shall retrieve and display recall data from the external API within five seconds of a user navigating to the Recalls page.
- The system shall return UPC product lookup results from the external product information data source specified in section 3.5.1 within three seconds of a valid UPC submission.

3.4. Assumptions and Restraints

3.4.1. **Assumptions (O: Josh)**

- Users shall have a reliable internet connection while using the application, as core functionality including recall data retrieval and UPC lookup depends on live API access.
- The FDA and USDA recall data made available through their public APIs shall be assumed accurate and current.
- The external product information data source specified in section 3.5.1 shall be assumed to contain product records for the majority of UPC codes that users are likely to scan.
- Users shall be assumed to maintain an accurate pantry by adding and removing items as their inventory changes, as the notification system depends on up-to-date pantry data.
- Users shall be assumed to grant browser notification permissions in order to receive real-time recall alerts.

3.4.2. **Restraints (O: Josh)**

- The system shall be developed and delivered with a single academic semester, limiting the scope of features to those defined in this specification.
- The system shall be limited to recall data available through the external government recall data data source specified in section 3.5.1. Recalls not present in this data source will not be reflected in the application.
- The system shall be limited to product lookups supported by the external product information data source specified in section 3.5.1. Products not present in that data source cannot be added to a user's pantry by UPC.
- The system shall be deployed as a prototype and is not intended to support production-scale traffic beyond the minimum concurrent user requirement defined in section 3.3.1..
- The system shall be constrained to the development tools and software dependencies specified in section 3.5.1.

3.5. Design Constraints

3.5.1. Software Dependencies (O: Parker D., M: Joshua A.)

- The system shall require NodeJS for backend development and server-side application logic.
- The system shall require SQLite3 for prototype database storage.
- The system shall require openFDA API as the external government recall data source for retrieving official food recall information.
- The system shall require OpenFoodFacts API as the external product information data source for retrieving product metadata by UPC.

- The system shall require an SMTP-compatible email service to send account verification and recall notification emails
- The system shall require internet connectivity to access external recall, product information, browser notification, and email delivery services.

3.5.2. Device Constraints (O: Parker D.)

- The user must have access to a browser and an internet connection to utilize the application.